

11. A hydrated compound $M_aAl_bZ_c \cdot dH_2O$ contains an s-block metal ion, M, and a halide ion, Z. The following tests were undertaken.

Test	Result
Heating to constant mass	A 0.0200 mol sample lost 1.44 g when heated
Adding concentrated sulfuric acid	Observations included coloured fumes, an orange-brown solution, steamy fumes and a choking gas; there was no smell of rotten eggs
Adding excess silver nitrate to a solution	A 25.0 cm ³ sample of a solution of concentration 0.203 mol dm ⁻³ produced a precipitate with a dry mass of 3.814 g
Adding excess sodium hydroxide to a solution	A white precipitate formed which dissolved in excess sodium hydroxide
Elemental analysis	The hydrated compound was found to contain 1.63 % by mass of s-block metal and 6.34 % by mass of aluminium

- (a) Calculate the value of d in the formula $M_aAl_bZ_c \cdot dH_2O$. [2]

$d =$

- (b) (i) Identify the halide present in the compound, explaining fully all observations that led to your conclusion. [2]

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(ii) Calculate the value of c in the formula $M_aAl_bZ_c \cdot dH_2O$.

[2]

$c =$

(c) State what the observations with sodium hydroxide indicate about the acidity/basicity of the aluminium ions. [1]

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(d) The M_r of the original **hydrated** compound is 425.62. Use all the information above to find the formula of the hydrated compound. Explain your reasoning. [3]

Formula of the **hydrated** compound

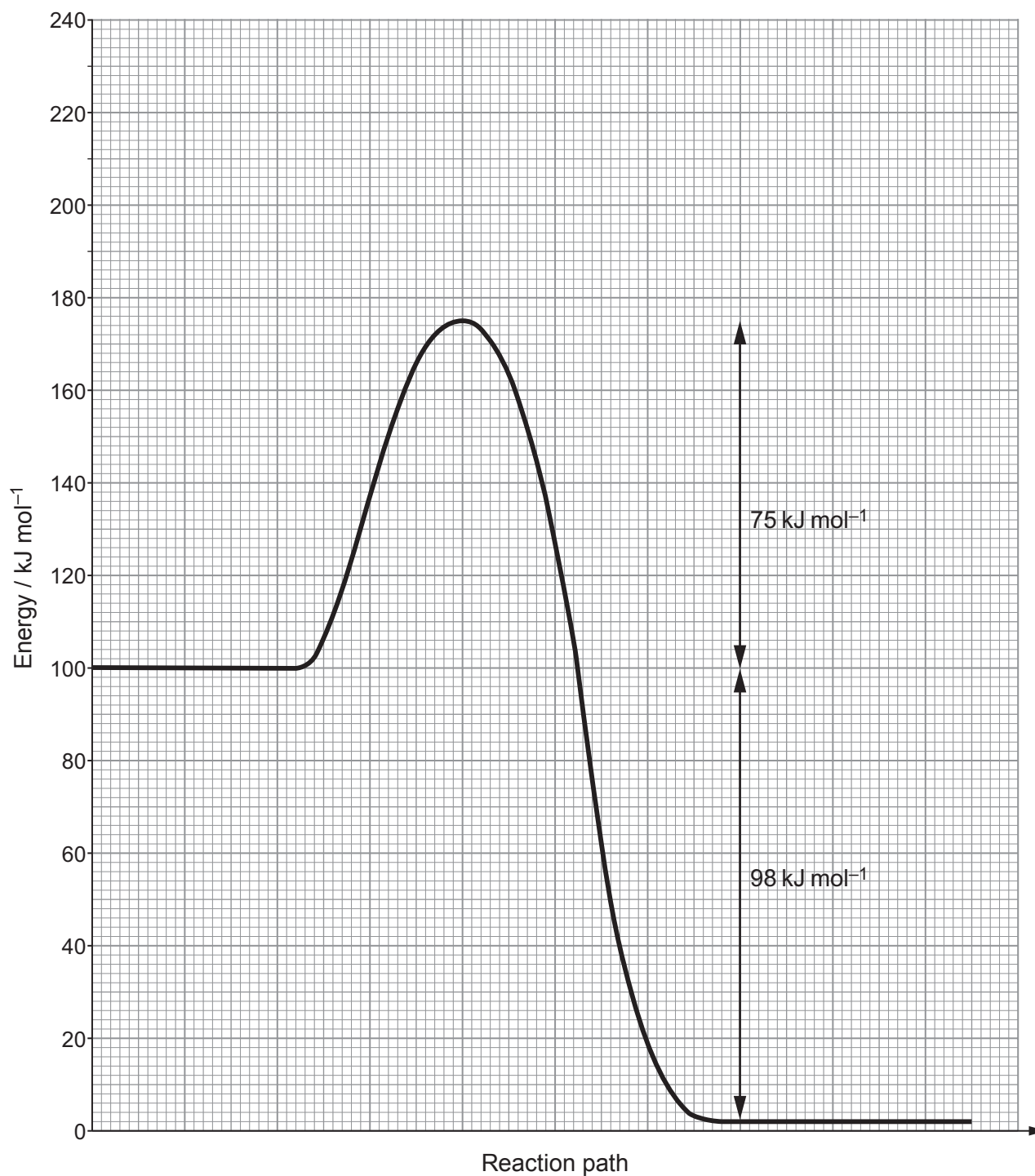
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11. Hydrogen peroxide, H_2O_2 , decomposes slowly at 300 K to form water and oxygen gas.



The reaction profile for this reaction is shown below.



- (a) The rate of decomposition of hydrogen peroxide can be influenced by a range of catalysts. The activation energy when using MnO_2 as a catalyst is 58 kJ mol^{-1} .

Draw the reaction profile for the catalysed reaction **on the grid opposite**. [1]

- (b) Another catalyst was used and this gave a value for the rate constant, k , of $1.68 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ and the frequency factor, A , of $1.41 \times 10^4 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ at a temperature of 300 K.

- (i) State the Arrhenius equation. [1]

- (ii) Calculate the activation energy using this catalyst and hence state whether it is a more effective catalyst than MnO_2 . [3]

Activation energy = kJ mol^{-1}

- (c) The standard enthalpy change of formation, $\Delta_f H^\theta$, of water is -286 kJ mol^{-1} .

Use this information and the graph to calculate the standard enthalpy change of formation of hydrogen peroxide. [2]

$\Delta_f H^\theta(\text{H}_2\text{O}_2) = \text{.....} \text{ kJ mol}^{-1}$

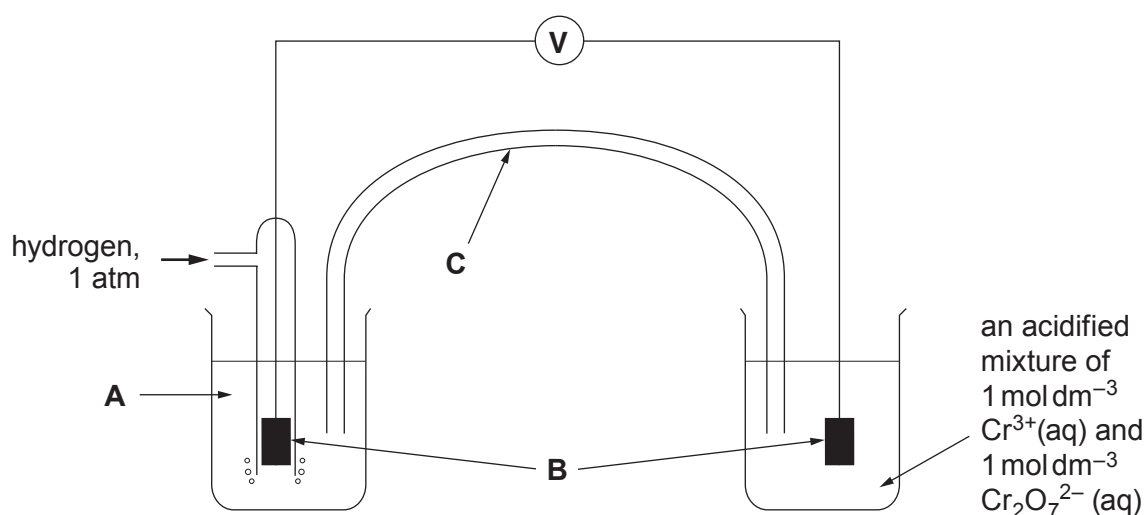
- (d) State whether you would expect the entropy change for the decomposition of hydrogen peroxide to be positive or negative. Give a reason for your answer. [1]



(e) One way of assessing whether a reaction is feasible is to use standard electrode potentials.

	Standard electrode potential / V
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.77
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1.33

(i) The apparatus below can be used to measure the standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$ half-cell.



I. State what is represented by **A** and **B** on the diagram.

[1]

A

B

II. **On the diagram** show the direction of flow of electrons in the external circuit.

[1]



III. State what is represented by **C** on the diagram and state its function. [1]

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IV. The concentrations of both $\text{Cr}_2\text{O}_7^{2-}$ and Cr^{3+} ions are 1 mol dm^{-3} . State and explain how the value shown on the high resistance voltmeter would change if the concentration of the Cr^{3+} ions were increased whilst the concentration of the $\text{Cr}_2\text{O}_7^{2-}$ was left unchanged. [2]

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(ii) It is suggested that hydrogen peroxide could be used to oxidise Cr^{3+} ions in acidic solution to form dichromate ions.

I. Write an equation for this proposed reaction. [1]

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II. Use the standard electrode potential values given to predict whether this reaction is feasible. [2]

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III. Another method of finding whether a reaction is feasible is to use the Gibbs free energy calculated from standard enthalpy of formation and standard entropy values.

State, giving a reason, whether Gibbs free energies or electrochemical methods are more appropriate for finding whether this reaction is feasible. [2]

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11. Oxalic acid is the common name for ethanedioic acid. It is found in a wide range of plants and when purified it can form a dihydrate, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$. The concentration of a saturated solution of oxalic acid can be found by titration.

(a) A sample of 25.0 cm^3 of a saturated solution of oxalic acid is diluted to form 250.0 cm^3 of an aqueous solution. Samples of 25.0 cm^3 of this diluted solution were heated and titrated using acidified potassium manganate(VII) to oxidise the oxalic acid.

(i) The half-equation for the oxidation of oxalic acid is shown below.



State the half-equation for the reduction of manganate(VII) ions in acid solution and use it to write the equation for the oxidation of oxalic acid by acidified manganate(VII) ions. [2]

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- (ii) The titration of 25.0 cm^3 samples of the diluted oxalic acid solution against a potassium manganate(VII) solution of concentration $0.0505\text{ mol dm}^{-3}$ gave the following results.

	1	2	3	4
Volume of KMnO_4 solution / cm^3	31.65	31.30	31.60	31.75

- I. Use appropriate data to find the mean volume of KMnO_4 solution used. [1]

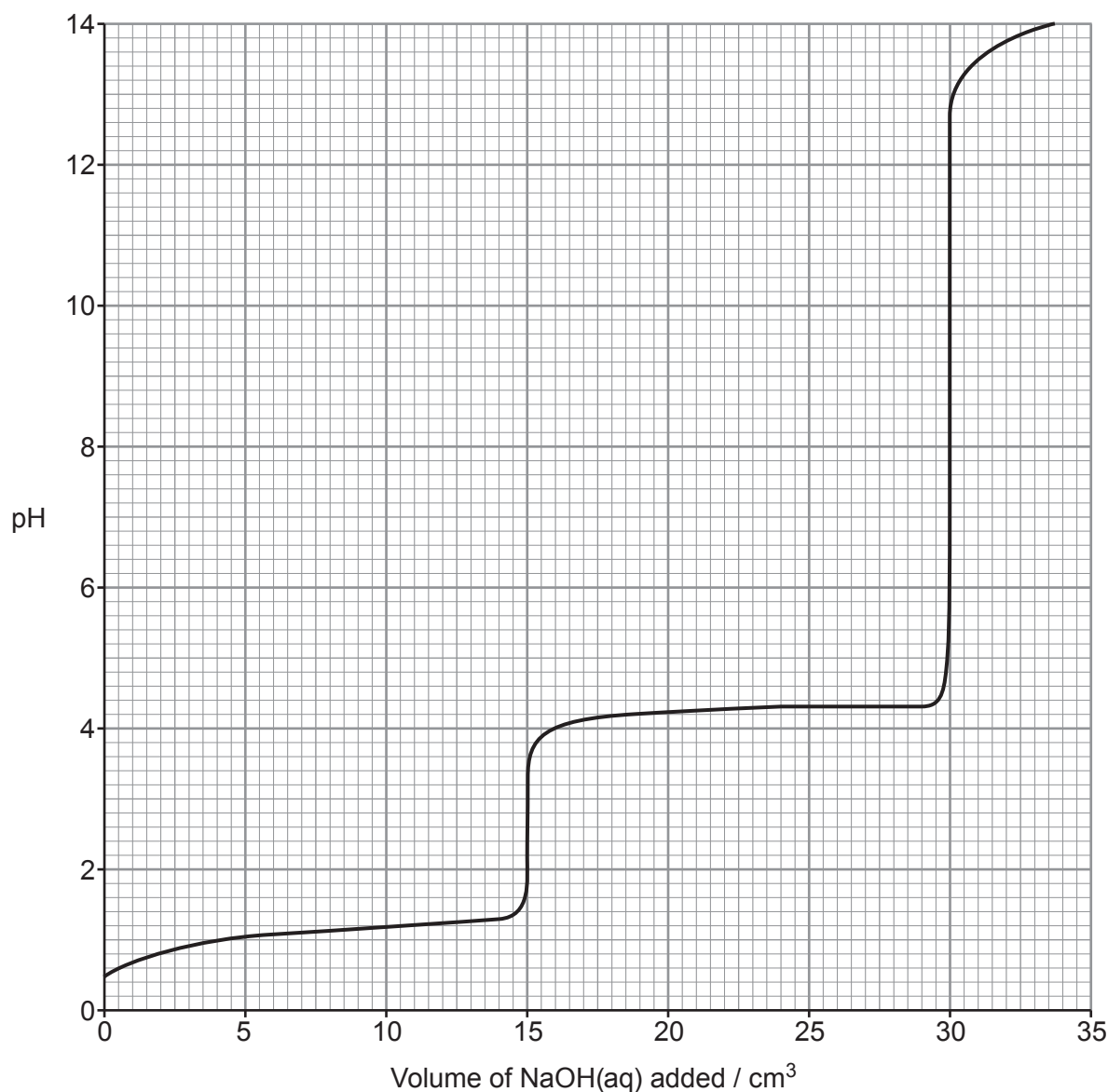
Mean volume = cm^3

- II. Calculate the concentration of the original saturated solution of oxalic acid. [3]

Concentration = mol dm^{-3}



- (b) Oxalic acid solutions may be analysed using acid-base titrations. The simplified pH curve for the titration of a solution of oxalic acid against sodium hydroxide is given below.



Four students undertook the titration in different ways:

- Alice used a pH probe to measure the pH after addition of each portion of sodium hydroxide solution
- Brychan used the indicator phenolphthalein in his titration; phenolphthalein has a pH range of 8.2 to 10.0
- Carys used the indicator methyl orange in her titration; methyl orange has a pH range of 3.2 to 4.4
- David used the indicator cresol red in his titration; cresol red has a pH range of 1.8 to 2.8



State and explain which, if any, of the methods chosen will allow the students to find valid data to calculate the concentration of oxalic acid. Briefly explain why oxalic acid has a pH curve of this shape. [6 QER]

You are not required to carry out any calculations.

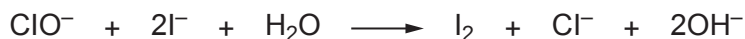
Examiner
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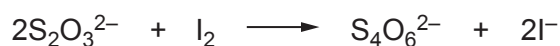


11. Chlorine can form a range of oxyanions with chlorine present in oxidation states including +1, +5 and +7.

- (a) Chlorate(I) ions, ClO^- , are commonly present in household bleaches. These ions can oxidise iodide ions to iodine.



Dewi took a 25.0 cm^3 sample of a household bleach and diluted it to form 250 cm^3 of solution in a standard volumetric flask. Four 25.0 cm^3 portions of the diluted solution were measured and placed in separate conical flasks and a spatula of solid potassium iodide was added to each. Each portion was titrated against standard sodium thiosulfate solution from a burette.



Dewi had access to five concentrations of sodium thiosulfate: 2.00 mol dm^{-3} , 1.00 mol dm^{-3} , 0.500 mol dm^{-3} , 0.200 mol dm^{-3} and $0.0500\text{ mol dm}^{-3}$.

He selected the 0.500 mol dm^{-3} solution and his results are shown in the table.

	1	2	3	4
Volume of $\text{Na}_2\text{S}_2\text{O}_3$ solution / cm^3	6.45	6.40	6.50	6.45

- (i) Calculate the mean volume of sodium thiosulfate solution used.

[1]

Mean volume = cm^3



- (ii) Find the concentration of sodium chlorate(I) in the initial bleach sample. [3]

Concentration = mol dm^{-3}

- (iii) Bleach concentrations are often quoted on labels as % w/v, that is the mass of sodium chlorate(I) dissolved in 100 cm^3 of water.

Calculate the % w/v concentration of sodium chlorate(I) in this bleach. [1]

Concentration = % w/v

- (iv) The teacher tells Dewi that he has selected an inappropriate concentration of sodium thiosulfate solution.

Suggest which concentration of sodium thiosulfate he should have chosen.
Give **two** reasons for your answer. [2]

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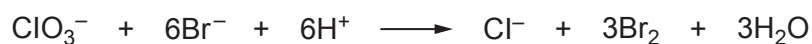
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- (b) Chlorate(V) anions, ClO_3^- , are strong oxidising agents and can oxidise bromide ions in acid solution to form bromine and chloride ions.



The initial rate of this reaction was measured at a temperature of 298 K using different concentrations of reactants.

Experiment	Concentration of ClO_3^- / mol dm^{-3}	Concentration of Br^- / mol dm^{-3}	pH	Initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.30	0.20	0	3.06×10^{-7}
2	0.60	0.20	0	6.12×10^{-7}
3	0.30	0.60	0	9.18×10^{-7}
4	0.30	0.60	1	

- (i) I. Find the orders of reaction with respect to chlorate(V) and bromide ions. [2]

You **must** show your working.

Order with respect to chlorate(V) ions

Order with respect to bromide ions



- II. The student finds that the rate of reaction is proportional to $[H^+]^3$. This is described as third order with respect to hydrogen ions.

Find the expected initial rate for experiment 4.

[2]

Initial rate = $\text{mol dm}^{-3} \text{s}^{-1}$

- (ii) Chemists often use an approximate rule that the rate of a reaction doubles when the temperature is increased by 10°C .

Show that this rule is true if this reaction is warmed from 298 K to 308 K. The activation energy for the reaction is 52.8 kJ mol^{-1} .

[3]

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- (c) The acid formed from chlorate(VII) ions is commonly called perchloric acid, HClO_4 . It is a very strong acid and is commonly classed as a superacid as it is a stronger acid than sulfuric acid.

- (i) State how the K_a value of a stronger acid will compare with that of a weaker acid, giving a reason. [1]

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- (ii) The concentration of 25.0 cm^3 of an aqueous solution of perchloric acid can be found by adding an excess of calcium carbonate and then measuring the amount of carbon dioxide gas released.

This can be done by either:

- measuring the volume of the gas released using a gas syringe, or
- measuring the mass lost as the carbon dioxide is released by placing the reaction flask on a digital balance.

Joe uses the first method to study his sample of perchloric acid and finds that the reaction releases 87 cm^3 of carbon dioxide at 30°C and 1 atm pressure.

Heledd uses the second method and finds that the mixture loses a total of 0.1533 g during the reaction.



- I. Find the number of moles of carbon dioxide released in each experiment and hence state whether the two solutions provided to Joe and Heledd are of the same concentration. [4]

You **must** show your working and give your answers to **appropriate** numbers of significant figures.

Moles of CO₂ using Joe's method = mol

Moles of CO₂ using Heledd's method = mol

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- II. Suggest, from the data provided, which of the two methods will give better results. Give a reason for your suggestion. [1]

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- III. Isha suggests using excess magnesium in place of calcium carbonate for Joe and Heledd's experiments. This will produce hydrogen gas in place of carbon dioxide. Explain which of the two methods will suffer the greater loss in accuracy. [2]

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- (iii) One common salt of perchloric acid is ammonium perchlorate.

Suggest a pH value for a solution of ammonium perchlorate. Give a reason for your answer. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
First name(s)		2

**GCE A LEVEL**

1410U30-1



S23-1410U30-1

MONDAY, 12 JUNE 2023 – MORNING

CHEMISTRY – A2 unit 3
Physical and Inorganic Chemistry

1 hour 45 minutes

Section A

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 7.	10	
8.	14	
9.	13	
10.	18	
11.	11	
12.	14	
Total	80	

1410U301
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.

Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q9(a)**.



JUN231410U30101

11. Carboxylic acids are classed as weak acids.

(a) State what is meant by the term weak acid.

[1]

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(b) Substituted propanoic acids, $\text{CH}_3\text{CHXCOOH}$, have different acid strengths. The table below gives some information about three substituted propanoic acids.

Substituted propanoic acid, $\text{CH}_3\text{CHXCOOH}$	
$\text{X} = \text{H}$	The pH of a 0.50 mol dm^{-3} aqueous solution is 2.59
$\text{X} = \text{Cl}$	$K_a = 1.48 \times 10^{-3} \text{ mol dm}^{-3}$
$\text{X} = \text{OH}$	$\text{p}K_a = 3.86$

Place these acids in order of increasing acid strength. **You must show your working.**

[3]

Weakest Strongest



- (c) A buffer is produced by mixing 100 cm^3 of ethanoic acid ($K_a = 1.8 \times 10^{-5}\text{ mol dm}^{-3}$) of concentration 0.80 mol dm^{-3} with 50 cm^3 of aqueous sodium hydroxide of concentration 0.80 mol dm^{-3} at a temperature of 298 K .

- (i) Calculate the initial pH of the aqueous sodium hydroxide. [2]

pH =

- (ii) Calculate the pH of the buffer solution. [2]

pH =

- (iii) Explain how buffers such as this work. [3]

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